

DNS & DHCP

1. Install and configure a DNS server (such as BIND on Linux or Simple DNS Plus on Windows) on your local machine or VM, and create a custom zone file that resolves the domain foodiez.in to the IP address 192.168.1.50.

->Here is a simple **BIND DNS Server** setup on Ubuntu/Linux that resolves foodiez.in to 192.168.1.50.

1. Install BIND

```
sudo apt update
```

```
sudo apt install bind9 bind9utils bind9-doc -y
```

Check that BIND is running:

```
sudo systemctl status bind9
```

2. Create the DNS Zone

Edit the BIND configuration:

```
sudo nano /etc/bind/named.conf.local
```

Add:

```
zone "foodiez.in" {  
    type master;  
    file "/etc/bind/db.foodiez.in";  
};
```

Save the file.

3. Create the Zone File

Create a new zone file:

```
sudo nano /etc/bind/db.foodiez.in
```

Enter:

```
$TTL 86400
```

```
@    IN    SOA    ns1.foodiez.in. admin.foodiez.in. (
      2026092701
      3600
      1800
      604800
      86400 )
```

```
@    IN    NS     ns1.foodiez.in.
```

```
ns1   IN    A     192.168.1.50
```

```
@    IN    A     192.168.1.50
```

```
www   IN    A     192.168.1.50
```

The important record is:

```
@    IN    A     192.168.1.50
```

This makes **foodiez.in** → **192.168.1.50**.

4. Check the Configuration

Run:

```
sudo named-checkconf
```

Then check the zone:

```
sudo named-checkzone foodiez.in /etc/bind/db.foodiez.in
```

You should see something similar to:

```
zone foodiez.in/IN: loaded serial 2026092701
```

OK

5. Restart BIND

```
sudo systemctl restart bind9
```

Enable it at startup:

```
sudo systemctl enable bind9
```

6. Test DNS Resolution

Install DNS utilities if necessary:

```
sudo apt install dnsutils -y
```

Then run:

```
dig @127.0.0.1 foodiez.in
```

You should find an answer containing:

```
foodiez.in. 86400 IN A 192.168.1.50
```

You can also test with:

```
nslookup foodiez.in 127.0.0.1
```

Expected Result

Domain	Record	IP Address
foodiez.in	A	192.168.1.50
www.foodiez.in	A	192.168.1.50
ns1.foodiez.in	A	192.168.1.50

Conclusion: The BIND DNS server is configured with a custom foodiez.in zone, and DNS queries for foodiez.in resolve to **192.168.1.50**.

2. Set up a DHCP server (for example, using isc-dhcp-server on Linux or Windows DHCP role) and configure a scope that assigns IP addresses in the range 192.168.1.100 to 192.168.1.150 with a lease time of 2 hours.

->DHCP Server Setup Using isc-dhcp-server on Linux

This example configures a DHCP scope that automatically assigns addresses from **192.168.1.100 to 192.168.1.150** with a **2-hour lease time**.

1. Install the DHCP server

On Ubuntu/Debian:

```
sudo apt update
```

```
sudo apt install isc-dhcp-server -y
```

2. Configure the DHCP scope

Open the DHCP configuration file:

```
sudo nano /etc/dhcp/dhcpd.conf
```

Add:

```
default-lease-time 7200;
```

```
max-lease-time 7200;
```

```
subnet 192.168.1.0 netmask 255.255.255.0 {  
    range 192.168.1.100 192.168.1.150;  
    option subnet-mask 255.255.255.0;  
    option routers 192.168.1.1;  
    option domain-name-servers 192.168.1.1;  
}
```

Why 7200?

2 hours × 60 minutes × 60 seconds = **7200 seconds**.

3. Select the network interface

Edit:

```
sudo nano /etc/default/isc-dhcp-server
```

Set your network interface, for example:

```
INTERFACESv4="ens33"
```

Replace ens33 with your actual interface name. You can find it with:

```
ip addr
```

4. Check the configuration

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
```

If there are no errors, restart the service:

```
sudo systemctl restart isc-dhcp-server
```

Enable it at startup:

```
sudo systemctl enable isc-dhcp-server
```

Check its status:

```
sudo systemctl status isc-dhcp-server
```

5. Verify the DHCP Scope

The configured range is:

Setting	Value
Network	192.168.1.0/24
DHCP Start	192.168.1.100
DHCP End	192.168.1.150
Available addresses	51
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
Lease time	2 hours (7200 seconds)

Expected Result

When a client connects to the network and requests an IP address, the DHCP server can assign an address such as:

192.168.1.100

192.168.1.101

192.168.1.102

...

192.168.1.150

Each DHCP lease will have a **2-hour lease time**.

3. Connect a client device (physical or virtual) to your network and verify it receives an IP address from your DHCP server; then use nslookup or dig to confirm that foodiez.in resolves to the correct IP address from your DNS server.

->DHCP Server Setup Using isc-dhcp-server on Linux

This example configures a DHCP scope that automatically assigns addresses from **192.168.1.100 to 192.168.1.150** with a **2-hour lease time**.

1. Install the DHCP server

On Ubuntu/Debian:

```
sudo apt update
```

```
sudo apt install isc-dhcp-server -y
```

2. Configure the DHCP scope

Open the DHCP configuration file:

```
sudo nano /etc/dhcp/dhcpd.conf
```

Add:

```
default-lease-time 7200;
```

```
max-lease-time 7200;
```

```
subnet 192.168.1.0 netmask 255.255.255.0 {
```

```
    range 192.168.1.100 192.168.1.150;
```

```
    option subnet-mask 255.255.255.0;
```

```
    option routers 192.168.1.1;
```

```
    option domain-name-servers 192.168.1.1;
```

```
}
```

Why 7200?

2 hours × 60 minutes × 60 seconds = **7200 seconds**.

3. Select the network interface

Edit:

```
sudo nano /etc/default/isc-dhcp-server
```

Set your network interface, for example:

```
INTERFACESv4="ens33"
```

Replace ens33 with your actual interface name. You can find it with:

```
ip addr
```

4. Check the configuration

```
sudo dhcpcd -t -cf /etc/dhcp/dhcpd.conf
```

If there are no errors, restart the service:

```
sudo systemctl restart isc-dhcp-server
```

Enable it at startup:

```
sudo systemctl enable isc-dhcp-server
```

Check its status:

```
sudo systemctl status isc-dhcp-server
```

5. Verify the DHCP Scope

The configured range is:

Setting	Value
Network	192.168.1.0/24
DHCP Start	192.168.1.100
DHCP End	192.168.1.150
Available addresses	51
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
Lease time	2 hours (7200 seconds)

Expected Result

When a client connects to the network and requests an IP address, the DHCP server can assign an address such as:

192.168.1.100

192.168.1.101

192.168.1.102

...

192.168.1.150

Each DHCP lease will have a **2-hour lease time**.

4. Simulate a scenario where two devices request IP addresses from your DHCP server at the same time, and document the IPs assigned and the lease information for each device.
Hint: Use the DHCP server's lease file or management console to view active leases and their expiration times.

->DHCP Lease Simulation

To simulate two devices requesting IP addresses at the same time, connect two DHCP clients to the network and set both to obtain an IP address automatically.

For an isc-dhcp-server on Linux, view active leases with:

```
sudo cat /var/lib/dhcp/dhcpd.leases
```

Example Result

Assume both devices requested an IP at **2:00 PM**.

Device	MAC Address	Assigned IP	Lease Start	Lease End	Lease Duration
Device 1 – Laptop	00:11:22:33:44:55	192.168.1.100	2:00 PM	4:00 PM	2 hours
Device 2 – PC	00:AA:BB:CC:DD:EE	192.168.1.101	2:00 PM	4:00 PM	2 hours

The DHCP server assigns different addresses from the configured pool:

192.168.1.100 – 192.168.1.150

Example Lease File Entries

The lease file will contain information similar to:


```
lease 192.168.1.100 {  
    starts 3 2026/09/27 14:00:00;  
    ends 3 2026/09/27 16:00:00;  
    hardware ethernet 00:11:22:33:44:55;  
    binding state active;  
}
```

```
lease 192.168.1.101 {  
    starts 3 2026/09/27 14:00:05;  
    ends 3 2026/09/27 16:00:05;  
    hardware ethernet 00:AA:BB:CC:DD:EE;  
    binding state active;  
}
```

Conclusion

Both devices successfully received **unique IP addresses** from the DHCP pool. Each address has a **2-hour lease**, and the lease file records the assigned IP, client MAC address, start time, expiration time, and active status.

5. Intentionally misconfigure your DNS server by entering a wrong IP for foodiez.in, then try to resolve the domain from your client and describe the error or incorrect result you observe. Finally, fix the error and verify that the resolution works correctly.

->DNS Misconfiguration and Troubleshooting

Using the BIND DNS server from the previous task, you can demonstrate this by temporarily changing the IP address for foodiez.in.

1. Intentionally enter the wrong IP

Open the zone file:

```
sudo nano /etc/bind/db.foodiez.in
```

Change:

```
@    IN    A    192.168.1.50
```

to an incorrect address, for example:

```
@    IN    A    192.168.1.99
```

Increase the serial number, for example:

```
2026092702
```

Then restart BIND:

```
sudo systemctl restart bind9
```

2. Test the DNS resolution

From the client, run:

```
nslookup foodiez.in
```

or:

```
dig foodiez.in
```

You would observe a result similar to:

```
Name:  foodiez.in
```

```
Address: 192.168.1.99
```

The important point is that **DNS resolution itself succeeds, but it returns the wrong IP address.** If no service exists at 192.168.1.99, trying to access the website would fail or reach the wrong host.

3. Fix the DNS record

Edit the zone file again:

```
sudo nano /etc/bind/db.foodiez.in
```

Change it back to:

```
@    IN    A    192.168.1.50
```

Increase the serial number again:

```
2026092703
```

Check the zone:

```
sudo named-checkzone foodiez.in /etc/bind/db.foodiez.in
```

Then restart BIND:

```
sudo systemctl restart bind9
```

4. Verify the corrected result

Run:

```
nslookup foodiez.in
```

Expected result:

Name: foodiez.in

Address: 192.168.1.50

You can also use:

```
dig foodiez.in
```

and confirm the **A record is 192.168.1.50**.

Conclusion

The intentional misconfiguration caused foodiez.in to resolve to the incorrect IP address **192.168.1.99**. After correcting the A record and reloading BIND, the domain correctly resolves to **192.168.1.50**. This demonstrates how an incorrect DNS record can cause clients to connect to the wrong server.